

Lecture 5: Risk and Term Structure of Interest Rates

Instructor: Fei Tan

 @econdojo  @BusinessSchool101  Saint Louis University

Course: Monetary Economics

Date: February 9, 2026

Overview

This lecture introduces three factors that affect bond yields: default risk, taxes, and maturity.

Key focus:

- The relationship among bonds with the same risk characteristics but different maturities, called the **term structure of interest rates**
- Two main explanations for the term structure
- Information contained in the risk structure and term structure of interest rates

The Road Ahead

1. Risk Structure of Interest Rates
2. Tax Status and Bond Types
3. Term Structure of Interest Rates
4. Information Content of Interest Rates

Risk Structure of Interest Rates

Everything else held equal, a ratings downgrade signifies a higher risk of default, shifting the bond demand curve to the left → Bond price falls, bond yield rises.

Any bond yield can be decomposed as follows:

$$\text{bond yield} = \underbrace{\text{U.S. Treasury yield}}_{\text{benchmark}} + \underbrace{\text{default risk premium}}_{\text{risk spread}}$$

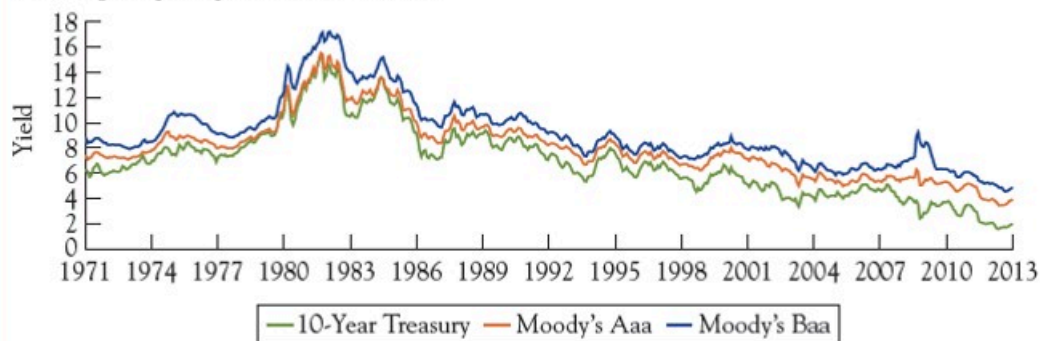
Key insight: When Treasury yields move, all other yields move with them.

Risk Structure of Interest Rates (cont'd)

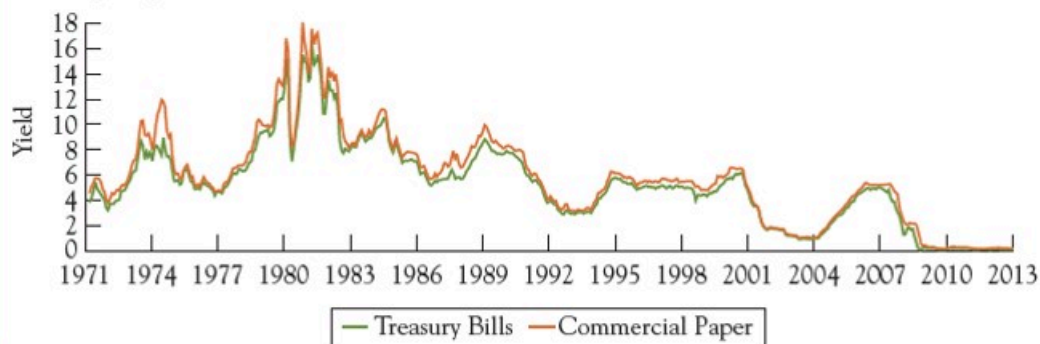
Figure 7.2

The Risk Structure of Interest Rates

A. Comparing Long-Term Interest Rates



B. Comparing Short-Term Interest Rates



Tax Status and Bond Types

Taxable bonds:

- Privately issued bonds
- Bondholders must pay income tax on interest income

Municipal (tax-exempt) bonds:

- Issued by state and local governments
- Coupon payments are exempt from federal taxation

$$\text{tax-exempt bond yield} = \text{taxable bond yield} \times (1 - \text{tax rate})$$

Example: With a 30% tax rate and 10% taxable bond yield, tax-exempt bond yield is $10\% \times (1 - 0.30) = 7\%$

Key insight: The higher the tax rate, the wider the gap between the two yields. Investors base their decisions on the tax-exempt yield.

Term Structure of Interest Rates

The term structure of interest rates is the relationship among bonds with the same risk characteristics but different maturities.

Three key observations:

1. Interest rates of different maturities tend to move together
2. Yields on short-term bonds are more volatile than yields on long-term bonds
3. Long-term yields tend to be higher than short-term yields

Figure 7.3

The Term Structure of Treasury Interest Rates



Explanation 1: Expectations Hypothesis

The expectations hypothesis assumes there is no uncertainty about the future. Bonds of different maturities are perfect substitutes for each other.

Example: An investor with a two-year horizon has two equivalent strategies:

1. Invest \$1 in a two-year bond and hold to maturity
 - Interest rate: i_{2t} (2 = two years, t = current period)
 - Payoff in two years: $(1 + i_{2t})^2$ dollars
2. Invest \$1 in two one-year bonds sequentially
 - Current one-year rate: i_{1t}
 - Expected one-year rate next period: i_{1t+1}^e (e = expected)
 - Payoff in two years: $(1 + i_{1t})(1 + i_{1t+1}^e)$ dollars

Indifference Condition

Indifference between the two strategies means:

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1t+1}^e)$$

A useful approximation:

$$i_{2t} \approx \frac{i_{1t} + i_{1t+1}^e}{2} \Rightarrow i_{1t+1}^e \approx 2i_{2t} - i_{1t}$$

More generally, the interest rate on an n -year bond is approximately the average of n expected future one-year interest rates:

$$i_{nt} \approx \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \cdots + i_{1t+n-1}^e}{n}$$

Interpreting the Yield Curve

Upward-sloping yield curve:

- Long-term rate (i_{2t}) > short-term rate (i_{1t})
- Financial markets expect short-term rates to rise in the future

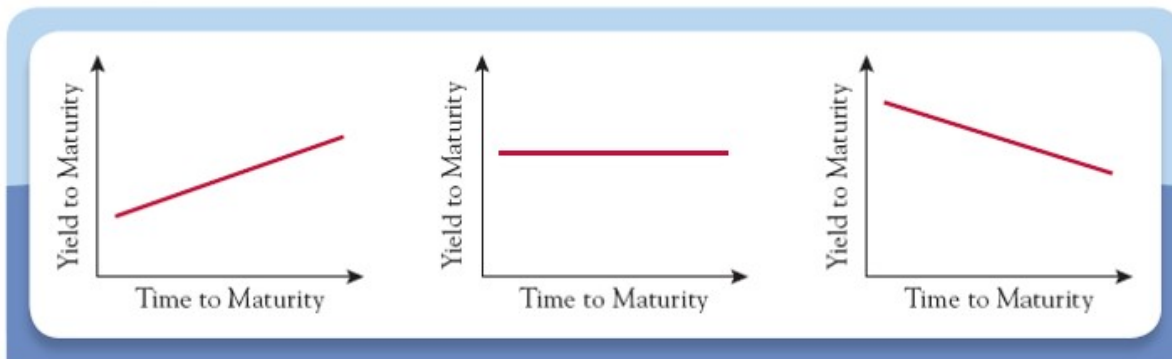
Downward-sloping yield curve:

- Long-term rate (i_{2t}) < short-term rate (i_{1t})
- Financial markets expect short-term rates to fall in the future

The yield curve shape reveals market expectations about future interest rates.

Figure 7.5

The Expectations Hypothesis and Expectations of Future Short-term Interest Rates



Can Expectations Hypothesis Explain Three Observations?

Observation 1: Interest rates of different maturities move together

- ✓ **Explained:** If i_{1t} changes, all yields at higher maturities change in the same direction

Observation 2: Short-term yields are more volatile than long-term yields

- ✓ **Explained:** Long-term yields are averages of expected future short-term yields, so they are less volatile

Observation 3: Long-term yields are normally higher than short-term yields

- ✗ **Not explained:** The yield curve slopes upward only when rates are expected to rise, but data shows rates have been trending downward

Explanation 2: Liquidity Premium Theory

The liquidity premium theory extends the expectations hypothesis to include risk.

Two components of bond yield:

1. **Risk-free rate** (explained by expectations hypothesis)
2. **Risk premium** (explained by inflation and interest-rate risk)

$$i_{nt} = rp_n + \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \cdots + i_{1t+n-1}^e}{n}$$

where rp_n is the risk premium for an n -year bond.

Risk Premium and Maturity

Key insight: The risk premium rp_n is an increasing function of maturity n .

Longer maturity → Higher inflation and interest rate risk → Higher risk premium

Implications:

1. Long-term yields are normally higher than short-term yields
2. Yield curve normally slopes upward
3. A flat curve means interest rates are expected to fall
4. A downward-sloping curve means interest rates are expected to decline significantly

Information Content of Interest Rates

The risk and term structure of interest rates contain valuable information for predicting overall economic conditions.

Two key indicators:

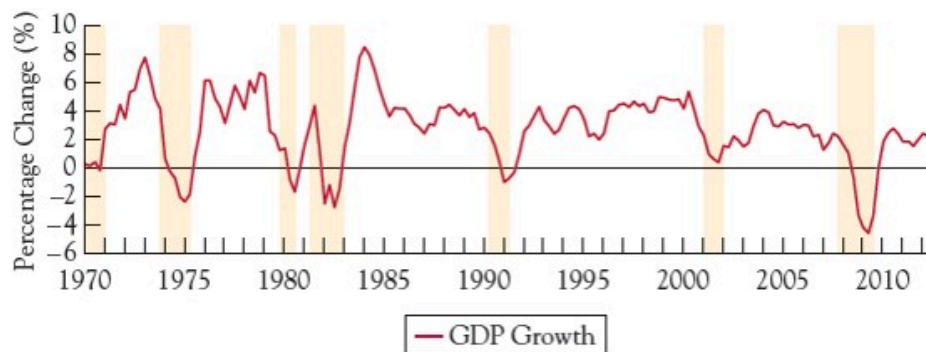
1. **Increasing risk spread** (difference between corporate and Treasury yields)
 - Signals deteriorating credit conditions
 - Predicts economic slowdown
2. **Inverted yield curve** (short-term rates $>$ long-term rates)
 - Historically precedes recessions
 - Reflects expectations of future economic weakness and declining interest rates

Risk Spread as an Economic Indicator

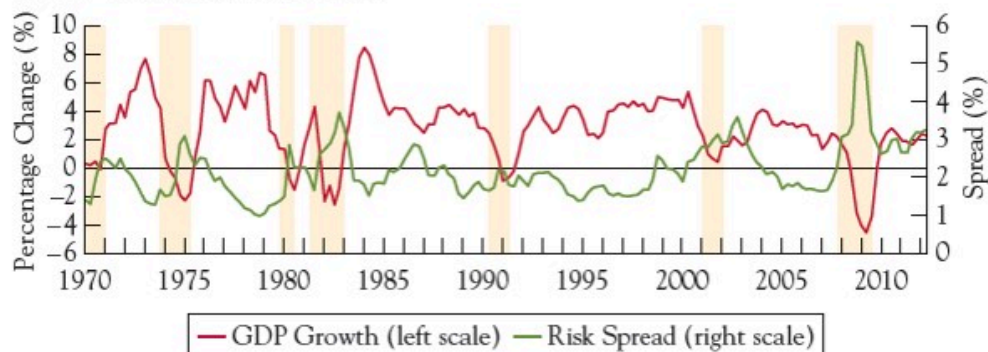
Figure 7.8

The Risk Spread and GDP Growth

A. GDP Growth with Recessions Shaded



B. GDP Growth with Risk Spread

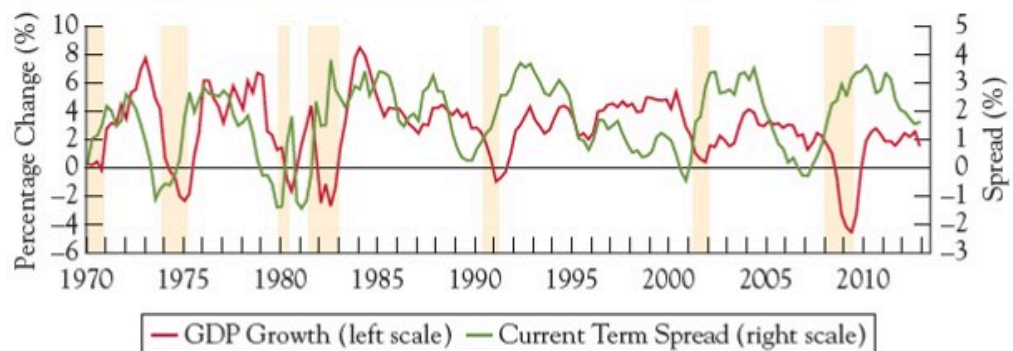


Yield Curve as an Economic Predictor

Figure 7.9

The Term Spread and GDP Growth

A. Current Term Spread and GDP Growth



B. GDP Growth with Term Spread 1 Year Earlier

